

Business Problem Statement

Flipkart, one of India's leading e-commerce marketplaces, manages an extensive catalog across various categories, brands, and sellers. To maintain a competitive edge, increase gross merchandise value (GMV), and improve overall buyer and seller satisfaction, management requires deeper insights into product pricing dynamics, customer demand drivers, seller ratings, and product performance.

Recent internal observations show high variance in sales volumes (`units_sold`) and customer satisfaction (`rating`) across core categories such as *Electronics*, *Mobiles*, *Fashion*, *Beauty*, *Appliances*, and *Home & Kitchen*. Management needs to understand how pricing, discount percentages, stock availability, seller ratings, and delivery terms directly influence purchasing decisions and return rates.

You are tasked with analyzing the Flipkart e-commerce dataset (`flipkart.csv` and `flipkart-cleaned.csv`) to answer the overarching business question:

"How can Flipkart leverage marketplace data to optimize pricing and discount strategies, improve seller performance, and boost product sales across key categories?"

Project Deliverables

1. Data Cleaning & Feature Preparation (Python)

- Ingest the raw dataset `flipkart.csv` and perform standard data cleaning techniques: handling missing values, standardizing column types, removing duplicates, and handling outliers.
- Derive key business features (such as actual revenue generated: $\text{final_price} \times \text{units_sold}$) and save the clean output as `flipkart-cleaned.csv`.

2. Analytical Querying & Segmentation (SQL)

Load `flipkart-cleaned.csv` and `flipkart.csv` into a relational SQL database. Execute SQL queries to answer critical business questions:

Top Revenue Drivers

Identify the top 10 products, categories, and brands by gross revenue.

Discount Elasticity

Analyze how varying levels of `discount_percent` impact total `units_sold`.

Seller Quality Metrics

Evaluate the relationship between `seller_rating`, `delivery_days`, and customer product ratings.

Product Return Dynamics

Determine which product characteristics correlate with high return rates (`is_returnable`).

3. Interactive Visualization & Executive Dashboard (Power BI)

Create an interactive Power BI dashboard referencing **flipkart-cleaned.csv** with the following views:

Executive Overview

Core KPIs including Total Sales Volume, Total Revenue, Average Discount Rate, and Average Product Rating.

Category & Brand Breakdown

Comparative analysis of performance across top brands (e.g., *Apple, Samsung, Puma, Dell*) and product categories.

Pricing & Discount Strategy

Scatter plots and correlation visuals showing price vs. sales volume across different discount tiers.

Seller & Logistics Insights

Visual indicators for seller ratings, delivery times, and inventory levels.

4. Executive Summary & Recommendations

Project Report

A structured documentation of your analytical methodology, key findings from **flipkart-cleaned.csv**, and strategic conclusions.

Presentation Deck

A visual slide deck delivering actionable recommendations for Flipkart management regarding pricing strategy, seller quality benchmarks, and inventory allocation.

5. GitHub Repository Structure

Maintain a organized GitHub repository containing:

 data/ flipkart.csv and flipkart-cleaned.csv	 notebooks/ Python/Jupyter scripts for data cleaning and EDA.	 sql/ Structured .sql scripts for analytical queries.
 dashboards/ Power BI (.pbix) report and dashboard visual exports.	 README.md Concise overview of project goals, key findings, and setup instructions.	